



LoP- The Logic of Propositions

Reasoning about what is True and what is False (HP2T)

LoP – The Logic of Propositions

- **Intuition**
- Definition
- Domain
- Language
- Interpretation function
- Theories
- Language entailment
- The meaning of logical connectives
- Logical entailment
- Reasoning problems
 - Model checking
 - Satisfiability
 - Validity
 - Unsatisfiability
 - Logical consequence
 - Logical equivalence
- Reasoning problems - correlations
- Key notions

LoP – Why a logic of propositions?

- The mainstream name for LoP is **PL** (for **Propositional Logic**);
- The **Logic of Propositions** (LoP) is the core (language) logic which allows to reason about propositions.
- It satisfies all the properties of logical entailment described above. In particular, it implements monotonic reasoning
- **All forms of monotonic reasoning on finite domains can be reduced to LoP reasoning**
- In particular, reasoning in all the world logics can be encoded in polynomial time into LoP reasoning
- In particular, Lol reasoning on finite domains can be encoded into LoP reasoning, paying the price of a worst case exponential blow-up of the reasoning time.
- All NP complete problems can be written in polynomial time as LoP problems

LoP – Highlights

- The version of LoP introduced here allows for all the main stream propositional connectives, that is:
 - Negation (not)
 - Conjunction (and)
 - Disjunction (or)
 - Implication (implies, if P then, $\rightarrow$).
 - Equivalence (iff)
 - Disjointness (exor)
- Many of this connectives could be eliminated by encoding them into others. The most common minimal sets of connectives (bases) are: and/or/not, implication/not

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LoP – Definition

We formally define the logic **LoP** as follows

$$\mathbf{LoP} = \langle \mathbf{L}_{\mathbf{LoP}}, \models_{\mathbf{LoP}} \rangle$$

where

$$\text{if } M \models_{\mathbf{LoDE}} a \text{ then } M \models_{\mathbf{LoP}} a'$$

with a and a' being in a one-to-one relation and

$$\mathbf{LoDE} = \langle \mathbf{EG}_{\mathbf{LoDE}}, \models_{\mathbf{LoDE}} \rangle$$

$$\mathbf{EG}_{\mathbf{LoDE}} = \langle \mathbf{L}_{\mathbf{LoDE}}, \mathbf{D}_{\mathbf{LoDE}}, \mathbf{I}_{\mathbf{LoDE}} \rangle$$

Below, any time no confusion arises, we drop the subscripts.

Observation (LoDE, LoP). LoDE is the Logic of Entity Bases, that is, of Entities together with their Definitions and Descriptions. LoP is the Logic of propositions (propositional logic).

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LoDE domain (Recall)

Definition (Domain, percepts)

$$D = \langle U, \{C\}, \{R\} \rangle$$

where:

- $U = \{u\}$ is called the **universe (of interpretation)** of D .
- $\{u\}$ is a set of **units** $u_1, \dots, u_n$, for some n
- $\{C\}$ is a set of **classes** $C_1, \dots, C_m$ of units, for some m , with $C_i \subseteq U$
- $\{R\}$ is a set of **binary relations** $R_1, \dots, R_p$ between units, for some p , with $R_i \subseteq U \times U$

Observation (EG, Binary relations). To comply to the graph notation we restrict ourselves to binary relations.



LoDE assertions – (Recall)

Observation. LoDE allows for the following assertions:

- Every etype (**primitive**, **defined** or **described**) / dtype and its argument is a fact.
- Every relation R and its two arguments is a fact.

Facts only have one of four possible forms:

- $E_T(e)$, meaning that the entity e is of etype E_T ,
- $D_T(v)$, meaning that the value v is of dtype D_T
- $O(e_1, e_2)$, meaning that the object property O holds between e_1 and e_2
- $A(e, v)$, meaning that the data property A of entity e has value v

Observation (LoE vs. LoDE). Differently from LoE, in LoDE we have not only primitive etypes, but also defined etypes and descriptions of etypes (in **bold**).

LoP Domain

Definition (Domain – facts).

$$D_{LoP} = \langle T, F \rangle$$

Definition (Domain – percepts). Let

$$D_{LoDE} = \langle E, \{C\}, \{R\} \rangle$$

be a LoDE domain of interpretation. Let $L_{LoDE} = \{a\}$ be a LoDE language for D_{LoP} where $\{a\}$ is the set of assertions in L_{LoDE} . Let $a_i \in L \subseteq L_{LoDE}$ be an assertion. Then

$$D_{LoP} = \{a^+, a^-\} = \{a_1^+, a_1^-, \dots, a_N^+, a_N^-\} = \{T(a_1), F(a_1), \dots, T(a_N), F(a_N)\}$$

where a^+, a^- are values of atomic propositions such that:

- $a^+ = T(a) = T$ if the LoDE assertion a is True
- $a^- = F(a) = T$ if the LoDE assertion a is False

Definition (Model). M is a set of atomic propositions $\{a^+, a^-\}$ such that, for each a , M contains one and only one between a^+ and a^- .

$$M = \{f\} = \{a^+, a^-\} = \{\dots, a_i^+, \dots, a_j^-, \dots\} \subseteq D_{LoP}.$$

Terminology (Model, atomic proposition). From now on, when no confusion arises, we talk of propositions meaning atomic propositions.

LoP domain - observations

Observation (Fact). As from the original intuition, a fact is what is the case in the world. In LoP, an analogical representation (model) is a set of judgements about what is true and what is false in the world. Therefore, in LoP the only two facts which can be observed are whether a certain judgement is true or whether a certain judgement is true false.

Observation (Percept). As from the original intuition, a percept is an element of the domain which can be perceived as distinct from other percepts. Despite that we only have two facts (Truth and Falsity) we have multiple percepts denoting one of the two facts. The distinguishing element of a percepts is the assertion being judged. One can think of propositions as being two disjoint sets of synonyms of one fact or the other.

Observation (A domain of propositions). $D = \{a^+, a^-\}$ as, if it allows for a judgement about an assertion, D also allows for the opposite judgement. This is because, as from the original intuition, a domain must allow for any fact to be case or not to be the case.

Observation (A domain of propositions). $a \in L \subseteq L_{LoDE}$ because LoP may be focused only on a subset of the assertions defined in LoDE.

Observation (Model). A LoP model has as many elements as there are propositions, that is, it is exactly half the size of the domain. For each proposition a model records its being true or false.

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Language (the same as LoE)

Definition (Assertional language)

$$L = \langle A, FR \rangle = \{P\}$$

where:

- L is a **propositional language**, where $P \in \{P\}$ is a **proposition**.
- A = is an **alphabet** of atomic propositions
- FR is a **set of formation rules**

Formation rules – BNF

$$\begin{aligned} \langle P \rangle & ::= \langle \text{atomic proposition} \rangle \mid \\ & \quad \neg \langle P \rangle \mid \\ & \quad \langle P \rangle \wedge \langle P \rangle \mid \\ & \quad \langle P \rangle \vee \langle P \rangle \mid \\ & \quad \langle P \rangle \supset \langle P \rangle \mid \\ & \quad \langle P \rangle \equiv \langle P \rangle \mid \\ & \quad \langle P \rangle \oplus \langle P \rangle \end{aligned}$$
$$\langle \text{atomic proposition} \rangle ::= P_1 \dots P_n \in \{P\}$$

Intuition

Observation (Proposition). We have four cases where a proposition P can assert **the truth or falsity of LoDE fact**, which in turn can be true /false. A proposition P , if true, depending on its content, asserts the falsity of a false fact or the truth or a truth fact

Observation (Negation). $\neg P$ asserts the opposite of the proposition P . It is the key connective for representing **negative information**.

Observation (Disjointness, exor). $P_1 \oplus P_2$, if true, asserts that the truth / falsity of one of the two propositions excludes the truth / falsity of the other. It is the key connective for representing **disjointness**.

Intuition

Observation (Conjunction). $P_1 \wedge P_2$, if true, asserts that both propositions are true. It is the key connective for transforming a set of propositions (about KG assertions) into a **single formula**.

Observation (Equivalence). $P_1 \equiv P_2$, if true, asserts that the truth / falsity of one of the two propositions guarantees the truth / falsity of the other. It is the connective for either **asserting equivalence** or defining a new proposition

Observation (Disjunction). $P_1 \vee P_2$, if true, asserts one or both propositions are true. It is the key connective for representing **partial information**

Observation (Implication). $P_1 \supset P_2$, if true, asserts that the truth of P_1 guarantees the truth of P_2 . It does not say anything about P_2 in case P_1 is false. It is the key connective for representing **logical consequence**.

Types of propositions

Intuition (Simple, articulative and nested propositions). In the description above of how to compute the truth value of (atomic) propositions we have assumed that propositions are judgements about LoDE assertions a , with $L_{\text{LoDE}} = \{a\}$. The implicit assumptions, not made explicitly is that these LoDE assertions were LoDE assertions, that is description of facts. We call these (atomic) propositions, **simple propositions**. However, this is not necessary the case. In fact we can also have propositions that express judgements about complex assertions made using the full power of the LoDE language, and also propositions about propositions. We call the first type of (atomic) propositions, **articulative propositions**, and the second **nested propositions**.

Intuition (Articulative propositions). These are propositions that make judgements about any LoDE assertion.

Example (Articulative propositions). (it is False that) “Silvia is a friend of paolo and she is 2 meters tall”.

Intuition (Nested propositions). We can have any level of nesting of propositions, that is propositions of propositions of ... assertions. This is extensively used in Natural languages. In logic, nested propositions are (sometimes) formalized using metatheories.

Example (Nested propositions). Propositions can be nested, assertions cannot. The nesting can also be “multiagent”.

- It is true that what Fausto said about the weather is false
- It is true that Mario said that what Fausto said about the weather is false

Observation (Problematic / interesting nested propositions).

- (It is true that) this sentence is false (**the Liar paradox**)
- (It is true that) this sentence is unprovable (**Goedel unprovable incompleteness formula**)

Note: In the following we only consider simple propositions.



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Interpretation function

Definition (LOP Interpretation function). Let $\{P\}$ the set of atomic proposition of a LoP language. Then

$$I_{\text{LoP}}: \{P\} \rightarrow \{T, F\}.$$

Observation (Truth value of an atomic proposition). In LoP how the truth value of an atomic proposition is computed is irrelevant.

Observation (Truth value of an atomic proposition). In LoP the truth value of an atomic proposition is computed is relevant from an application point of view as it is constrained by the world as described by a LoDE EG and constrains (when a truth value is imposed on an proposition whose LoDE truth value is undefined).

Interpretation function

Definition (From I_{LoP} to I_{LoDE}).

- Let $D = \langle E, \{C\}, \{R\} \rangle$ be a LoDE domain of interpretation.
- Let $L_W = \langle W, \models_{LoDE} \rangle$ be a LoDE Logic and $L_{LoDE} = \{a\}$ be a LoDE language for D where a is an assertion and $\{a\}$ is the set of assertions in L_{LoDE} .
- Let $L_{LoP} = \{P\}$ be LoP language, where $P \in \{P\}$ is a proposition.
- Let I_{LoP} be a LoP interpretation function, with $I_{LoP}: \{P\} \rightarrow \{T, F\}$.

Then we have

$$\text{If } \models_{LoDE} a \text{ then } I_{LoP}(a^+) = T \text{ and } I_{LoP}(a^-) = F \quad (*)$$

Observation (From I_{LoP} to I_{LoDE}). If $(*)$ does not hold then I_{LoP} is not constrained and it will assign a truth value depending on the LoP theory.

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Theory building

Intuition (Theory building). The theory building is performed in three steps

- **(Step 1):** Define the LoP reference model, that is, the set of LoDE assertions which describe the facts which are true in the model
- **(Step 2):** Define the LoP language, that is, the set of atomic propositions and logical connectives which are used to judge what is true / false in the model
- **(Step 3):** Define the LoP theory, that is, the set of (atomic and complex) propositions which constrain what is the case in the model by:
 - (1) specifying the negative knowledge,
 - (2) completing the partial information encoded by the model, and
 - (3) putting further constraints on what is the case via complex propositions. This is highly dependent on the reasoning task (see below)

Theory building (step 1)

Intuition (Define the LoP Reference Model). The first step is articulated in five phases:

- **(Phase 1a)** Define the set of LoE assertions of the EG
- **(Phase 1b)** Define the set of LoC/LoD language definitions
- **(Phase 1c)** Define the set of LoC/LoD knowledge descriptions
- **(Phase 1d)** Perform the LoC/LoD unfolding
- **(Phase 1e)** Perform the LoDe expansion

Observation (Define the LoP reference model). Any of the first three steps is optional. Step 1d and Step 1e are performed only when needed. The key observation is that LoP propositions can be built by expressing judgements on all three LoDE components: ground facts about entities, facts about defined etypes, facts about language concepts.

Theory building (step 2)

Intuition (Define the LoP Language). The second step is articulated in three phases:

- **(Phase 2a)** Select which LoDE assertions are going to be judged
- **(Phase 2b)** Select a uniform method for encoding a LoDE assertion a into a LoP assertion a'^+ , a'^- . This in turn is composed of two steps
 - (1) How to encode a structured formula into an atomic formula, e.g., from *HasFriend(Stefania#1,Paolo#1)* to *HF-S.P*
 - (2) which of the possible positive or negative encodings a'^+ , a'^- select and how to encode them in the proposition name, e.g., from *HF-S.P* to *HF-S.P0* and *HF-S.P1*
- **(Phase 2c)** Select the logical connectives, not necessarily used to write complex propositions

Intuition (Phase 2b). There is a std encoding which performs a 1-to-1 mapping (see later).

Theory building (step 3)

Intuition (Define the LoP Theory). The third step is articulated in three phases:

- **(Phase 3a).** Select the LoDE assertions which are going to be judged. This usually turns out to be a set of atomic or conjunctions of atomic propositions
- **(Phase 3b).** Select the negative knowledge, implicitly encoded in the LoDE theory, to be made explicit in the LoP theory. This usually turns out to be a set of negations, or disjointness or implication axioms.
- **(Phase 3c).** Select the partial knowledge, implicitly encoded in the LoDE theory, to be made explicit in the LoP theory. This usually turns out to be a set of disjunction axioms.
- **(Phase 3d).** Add a set of LoP axioms which encode the information provided linguistically which refines what is not explicitly stated in LoDE

Observation (Define the LoP theory). Usually, not all the implicit negative and partial knowledge of a LoD theory is made explicit in a LoP theory, in particular when it takes, implicitly or explicitly, the form of disjunctions. The reason being that the complexity of reasoning grows exponentially with the number of disjunctions.

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Language Entailment (Recall)

Notion 6 (Entailment) Let $W = \langle L_a, D, I_a \rangle$ be a world model. $L_W = \langle W, \models_{L_a} \rangle$ be a world logic for W . Let $L = \{w\}$ be a language, with $L_a \subseteq L$. Let $M = \{f\} \subseteq D$ be a set of facts. Let $T \subseteq L$ be a theory. Then $\models_L$ is an **entailment relation** that associates the facts in M with the elements in T , in formulas

$$\models_L \subseteq D \times L, \text{ also written } M \models_L T, (*)$$

subject to the constraint that for all assertions $a \in L_a$,

$$\text{if } M \models_{L_a} a \text{ then } M \models_L a \quad (**)$$

We also say that M **entails** T and write $M \models T$ when no confusion arises.

Observation ($L_a \subseteq L$). Usually the assertions $a \in L_a$ get rewritten, to assertions $a' \in L$, under the guarantee of a one-to-one mapping between the two notations.

Observation ($L_a \subseteq L$). $(**)$ holds only in one direction because of the partiality of world logic models. Remember that we have restricted ourselves to complete logic models.

Language entailment

$M \models P,$	if $I(P) = T,$	with	$P \in \{P\}$
$M \models \neg P,$	if		$M \not\models P$
$M \models P_1 \wedge P_2,$	if $M \models P_1$	and	$M \models P_2$
$M \models P_1 \vee P_2,$	if $M \models P_1$	or	$M \models P_2$
$M \models P_1 \supset P_2,$	if when $M \models P_1,$	then	$M \models P_2$
$M \models P_1 \equiv P_2,$	if $M \models P_1$	if and only if	$M \models P_2$
$M \models P_1 \oplus P_2,$	if $M \models P_1$	if and only if	$M \not\models P_2$

Model and interpretation

Proposition (Interpretation and model). An interpretation assigns a truth value to all the atomic formulas, in this case, the atomic propositions, of a language, in this case a LoP language. As from above, a model also assigns a truth value to all atomic propositions. That is, in LoP, models and interpretations are the same formal object.

Proposition (Interpretation and model). A model of a formula is an interpretation which entails that formula.

Notation (Model and interpretation). Being models interpretations, we write and say that interpretations entail formulas.

Language entailment

$I \models P,$	if $I(P) = \top,$	with	$P \in \{P\}$
$I \models \neg P,$	if		$I \not\models P$
$I \models P_1 \wedge P_2,$	if $I \models P_1$	and	$I \models P_2$
$I \models P_1 \vee P_2,$	if $I \models P_1$	or	$I \models P_2$
$I \models P_1 \supset P_2,$	if when $I \models P_1,$	then	$I \models P_2$
$I \models P_1 \equiv P_2,$	if $I \models P_1$	if and only if	$I \models P_2$
$I \models P_1 \oplus P_2,$	if $I \models P_1$	if and only if	$I \not\models P_2$

Interpretation equivalence wrt. a formula

Proposition (Interpretation equivalence wrt a formula). Let A be a formula where P is the set of all the atomic propositions which occur in it. If

$$I(P) = I'(P),$$

then

$$I \models A \text{ iff } I' \models A.$$

That is:

- The truth value of atomic propositions which occur in A fully determines the truth value of A
- The truth value of the atomic propositions which do not occur in A play no role in the computation of the truth value of A ;

Observation (Interpretation equivalence and model generation). The above proposition allows us to focus only on the propositions which occur in a formula. The interpretation of the other atomic propositions are irrelevant to deciding whether an interpretation is a model of a formula.

Model and theory, observations

Observation (Maximum number of models for a LOP language). If $|\{P\}|$ is the cardinality of $\{P\}$, then there are $2^{|\{P\}|}$ different models, corresponding to all the different subsets of $\{P\}$.

Observation (Number of theories for a model). A LOP model can be (partially) described by multiple theories, as long as these theories assign the same truth value as the model to any subset of the model propositions.

Observation (Number of models of a theory). A theory T has usually multiple models (which complete the assignment of truth values to all atomic propositions). T can have any number of models between 0 (when it contains a contradiction) and $2^{|\{P\}|}$ when all its formulas are tautologies.

Minimal models, maximal theories

Observation 14 (Partiality of LoP theories). The more partial a LoP theory T is, in terms of truth values assigned to propositions, the more models. For instance, assume $\{P\} = \{P_1, P_2\}$.

- $T = \{P_1 \vee \neg P_1, P_2 \vee \neg P_2\}$, has four models
- $T = \{P_1 \vee \neg P_2\}$, has three models
- $T = \{P_1\}$ has two models
- $T = \{P_1 \wedge P_2\}$ has one model
- $T = \{P_1 \wedge \neg P_1\}$ has no models

Definition (Maximal theory - reprise). A maximal theory is a theory which has only one model

Observation (Maximal theory - reprise). A model has multiple maximal theories. For instance, $T_1 = \{P_1, P_2\}$, and $T_2 = \{P_1 \wedge P_2\}$, with $\{P\} = \{P_1, P_2\}$ are two maximal theories for the same model.

Definition (Minimal model). Given a theory, a minimal model is a model which is the intersection of all the models for that theory.

Observation (Minimal model). Minimal models do not necessarily exist. For instance, assume $\{P\} = \{P_1, P_2\}$. $T = \{P_1 \vee P_2\}$ has three models and no minimal model.

Model as set of true propositions

Terminology (A Model as a set of true propositions). A model, can be represented set theoretically as the set of atomic propositions it defines as true.

Example (Model). See the table on the right.

Terminology (model). A model M can be thought as a subset S of $\{P\}$ where I , an interpretation function, is the characteristic function of S , i.e.

$A \in S$ if and only if $I(A) = \text{True}$.

	p	q	r	Set Theoretic Representation
I_1	True	True	True	$\{p, q, r\}$
I_2	True	True	False	$\{p, q\}$
I_3	True	False	True	$\{p, r\}$
I_4	True	False	False	$\{p\}$
I_5	False	True	True	$\{q, r\}$
I_6	False	True	False	$\{q\}$
I_7	False	False	True	$\{r\}$
I_8	False	False	False	$\{\}$

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Observation: How Connectives Operate

Negation	
$\neg$ True	False
$\neg$ False	True

Conjunction	
True $\wedge$ True	True
True $\wedge$ False	False
False $\wedge$ True	False
False $\wedge$ False	False

Disjunction	
True $\vee$ True	True
True $\vee$ False	True
False $\vee$ True	True
False $\vee$ False	False

Disjointness (exor)	
True $\oplus$ True	False
True $\oplus$ False	True
False $\oplus$ True	True
False $\oplus$ False	False

Equivalence	
True $\equiv$ True	True
True $\equiv$ False	False
False $\equiv$ True	False
False $\equiv$ False	True

Consequence	
True $\supset$ True	True
True $\supset$ False	False
False $\supset$ True	True
False $\supset$ False	True

Negation

Principle of non-contradiction

$$\neg (P \wedge \neg P)$$

- Never true, in no model.
- It imposes the key constraint for a model to be a possible picture of reality
- All formulas of the form $P \wedge \neg P$, independently of the shape of P , are called **contradictions**.
- Sometimes $P \wedge \neg P$ is written as $\perp$ (for falsity, as represented in the language). Not to be confused with $\perp$ (Bottom) in LoDE!
- The interpretation of $\perp$ is F

Principle of the excluded middle

$$P \vee \neg P$$

- True in all models.
- It imposes the key constraint for a model to be complete. P or $\neg P$ must be true in a model, but not both, for the law of non-contradiction
- All formulas of the above form, independently of the shape of P , are called **tautologies**.
- Sometimes $P \vee \neg P$ is written as $\mathbf{T}$ (for truth, as represented in the language). Not to be confused with $\mathbf{T}$ (Top) in LoDE!
- The interpretation of $\mathbf{T}$ is T

Observation (Negation). The meaning of negation is given by the two laws above. They are theorems of LoP. The first enforces in the language the fact that a model cannot contain two facts (and therefore two propositions) which contradict one another. The second enforces in the language that (1) a proposition is either True or False, and (2) that statement (1) is true.

Negation

Proposition 1 (Principle of non contradiction): $\neg (P \wedge \neg P)$ is a LoP tautology.

Proposition 2 (Disjointness): $P \oplus \neg P$ is a LoP tautology

Proposition 3 (Consequence): $P \supset \neg \neg P$ is a LoP tautology

Proposition 4 (Consequence): $\neg \neg P \supset P$ is a LoP tautology

Proposition 5 (Equivalence): $\neg \neg P \equiv P$ is a LoP tautology

Proposition 6 (Principle of the excluded middle): $P \vee \neg P$ is a LoP tautology

Negation

Observation (Representing negation). Tautologies (1), (2) are «more primitive» than tautologies (3) – (6) as they only depend on fact (1) in the previous page (that is, that a proposition is either true or false), while the others exploit also fact (2) in the previous page (namely that fact (1) is itself a fact).

Observation (Representing negation). Tautologies (1), (2) are useful in that they allow to infer, once one knows that one of the two elements is true (false) then the other is false (true). That is, they allow to distinguish negative knowledge from partial knowledge (that is, «knowing not» from «not knowing»). Tautologies (3)-6) allow additionally to prove the equivalence of the various tautologies, for instance, all the ones listed above. They enforce the use of truth tables.

Conjunction / disjunction / negation

Same proposition

$$A \wedge A \equiv A$$

$$A \vee A = A$$

Commutativity

$$A \wedge B \equiv B \wedge A$$

$$A \vee B \equiv B \vee A$$

De Morgan laws

$$\neg (A \vee B) \equiv \neg A \wedge \neg B$$

$$\neg (A \wedge B) \equiv \neg A \vee \neg B$$

Associativity

$$(A \wedge B) \wedge C \equiv A \wedge (B \wedge C)$$

$$(A \vee B) \vee C \equiv A \vee (B \vee C)$$

Distributivity

$$A \wedge (B \vee C) \equiv (A \wedge B) \vee (A \wedge C)$$

$$A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$$

Conjunction / disjunction / negation

Observation ($\neg$, $\wedge$, $\vee$ in set theory). The above formulas are LoP tautologies. They are mapped one-to-one with the fundamental operations and properties of sets, that is, complement, conjunction and negation, and the De Morgan laws. This is a key property which enforces the fact that reasoning about truth in LoP maps one-to-one with the set operations in set-theory. This is they in the grounding of LoP into LoDE

Observation ($\neg$, $\wedge$, $\vee$ as a LoP base). $\neg$, $\wedge$, $\vee$ are a base for LoP, meaning by this that all the other connectives can be defined from these three. This is a key property which enforces the facat that all the reasoning which can be done in LoP can be explained in terms of set-theoretic operations. Namely that, ultimately, we can reason linguistically about what we perceive in analogic representations.

Implication / equivalence – the meaning of Logical Equivalence

Implication and disjunction (1)

$$(A \supset B) \equiv (\neg A \vee B)$$

Implication and negation (2)

$$A \supset B \equiv \neg B \supset \neg A$$

Implication and contradiction (3)

$$\perp \supset A, \text{ for any } A$$

Implication and equivalence (4)

$$(A \equiv B) \equiv ((A \supset B) \wedge (B \supset A))$$

Equivalence and disjointness (5)

$$(A \equiv B) \equiv \neg(A \oplus B)$$

Disjointness and disjunction (6)

$$(A \oplus B) \equiv (\neg A \wedge B) \vee (A \wedge \neg B)$$

Implication / equivalence – the meaning of Logical Equivalence

Observation (Implication). The above formulas are LoP tautologies. **Implication is the LoP symbol which allows to describe how logical consequence works.**

- (1) says that if A is true then B must be true.
- (2) says that if B is false then A must be false (the counterpositive).
- (3) says that a contradiction (since it does not have models) allows to derive everything .
- (4) defines equivalence as computing the implication in both directions.
- (5) says the exor (disjointness) is the equivalence to two propositions which negate one another.

Implication / conjunction / disjunction / negation

Implication and conjunction (1)

$$(A \wedge B) \supset C \equiv (A \supset C) \vee (B \supset C)$$

Implication and conjunction (2)

$$(A \wedge B) \supset C \equiv A \supset (B \supset C)$$

Implication and conjunction (3)

$$(A \wedge B) \supset C \equiv A \supset (C \vee \neg B)$$

Implication and conjunction (4)

$$A \supset (B \wedge C) \equiv (A \supset B) \wedge (A \supset C)$$

Implication and disjunction (5)

$$(A \vee B) \supset C \equiv (A \supset C) \wedge (B \supset C)$$

Implication and disjunction (6)

$$(\neg A \vee B) \supset C \equiv (A \supset B) \supset C$$

Implication and disjunction (7)

$$(A \supset (B \vee C)) \equiv (A \wedge \neg B) \supset C$$

Implication and disjunction (8)

$$(A \supset (B \vee C)) \equiv (A \supset B) \vee (A \supset C)$$

Implication / conjunction / disjunction / negation

Observation (Conjunction and implication). The above formulas are LoP tautologies.

- Look at equation (1): conjunction in the premise of an implication weakens the implication (more requirements for the conclusion to be true).
- Look at equation (4): conjunction in the conclusion of an implication strengthens the implication when in the conclusion (more truths for the same output).

Observation (Disjunction and implication). The above formulas are LoP tautologies.

- Look at equation (5): disjunction in the premise of an implication strengthens the implication (more truths for the same output)
- Look at equation (8): disjunction in the conclusion of an implication weakens the implication when in the conclusion (more requirements for the conclusion to be true).

Observation (Conjunction vs Disjunction). Conjunction and disjunction show an opposite behavior. Beyond what written above, the pairs of tautologies (2), (6) and (3), (7) show their opposite behaviour with respect to nested implications.

LoP vs. LoD connectives – observations

Observation (LoP Negation, conjunction, disjunction vs LoD complement, intersection, union).

Because of the direct mapping between LoP and set-theory (see above), these three LoP propositional connectives are the main means for translation with the LoD connectives and composite etypes and LoP.

Observation (LoP vs LoD: equivalence, disjointness). LoP equivalence and disjointness map only partially to LoD equivalence and disjointness. In LoP, these symbols can be nested while this is not the case in LoD. This is a main advantage of moving from the LoD set-theoretic semantics to the LoP truth-theoretic semantics.

Observation (LoP implication vs LoD Subsumption). The same as for equivalence and disjointness.

Observation (LoP vs LoD nesting of operators). The LoD semantics do not allow for the nesting of subsumption, equivalence and disjointness, because, differently from negation (that is complement), conjunction (that is intersection) and disjunction (that is union), they do not construct a set but, rather, they return a truth value (as it is the case in LoP). LoP, since it only works on truth values, allows for the nesting of ALL logical connectives.

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Logical entailment (Recall) – the same as in world logics

Definition (Logical entailment). Let $L_W = \langle W, \models_W \rangle$ be a world logic with $W = \langle L, D, I \rangle$ its world model. Let $\{M\} \subseteq D$ be a set of models of T with $T, \underline{I} \subseteq L$ and $a \in L$. Then we write

$$T \models_{\{M\}} \underline{I} \qquad (T \models_{\{M\}} a)$$

and say that T **logically entails** $\underline{I}$ (T **logically entails** a) with respect to $\{M\}$ if

$$\text{for all } M \in \{M\}, \text{ if } M \models T \text{ then } M \models \underline{I} \text{ (} M \models a \text{)}$$

Observation (Model filtering via logical entailment). Logical entailment implements reasoning in the following way: “Whatever we derive from T must be true in the same sets of “relevant” models $\{M\}$ as T . T acts as a filter which allows to discard all the irrelevant models.

Observation (Model filtering via selection of $\{M\}$). The choice of the models $\{M\}$ of T is arbitrary. And it also amounts to model filtering. It could be the only model of T , all models of T , a model selected because it is the intended model of T , and so on. If $M \not\models T$ then Logical entailment does not apply. T has the key role of selecting which models $M \in \{M\}$ must be considered.

Notation ($\{M\}$). When no confusion arises we write $\models$ instead of $\models_{\{M\}}$.

Logical entailment – properties (P1) (same as world logics)

Definition (Reflexivity). Let w be any formula of some logical language. Then

$$w \models w$$

Observation (Reflexivity). Every fact entails itself. Knowledge asserts itself as being knowledge. This is the essence of what knowledge is about. That is, if a formula is true in a representation (the memory of what we have perceived or been told), then I'll answer so any time I am asked.

Observation (Reflexivity). Reflexivity, to hold, requires that one is capable of recognizing two occurrences of the same formula (proposition, assertion, fact) to be the same. That is (as stated earlier on) we need a theory of (dis)similarity.

Observation (axioms, theorems). Because of reflexivity all axioms are theorems.

Logical entailment – properties (P2) (same as world logics)

Definition (Cut). Let w_1, w_2 be any formulas of some logical language, T and Σ any set of formulas. Then

$$\text{If } T \models w_1 \text{ and } \Sigma \cup \{w_1\} \models w_2 \text{ then } T \cup \Sigma \models w_2$$

Observation (Cut) There are two ways to interpret cut.

The first and most common relates to **efficiency**. That is, reasoning can be made efficient by dropping intermediate irrelevant results.

The second relates to **transitivity**. That is, reasoning can be computed by chaining independent reasoning sessions, something that people do all the time during their everyday life.

Logical entailment – properties (P3) (same as world logics)

Definition (Compactness). Let w be any formula of some logical language, T any set of formulas. Then

If $T \models w$ then there is a finite subset $T_0 \subseteq T$ such that $T_0 \models w$

Observation (Compactness). Compactness says that we do not need to use all we know to derive our goal.

This is a property extensively used by humans in common sense reasoning. We always minimize the amount of knowledge we use when reasoning.

This applies also in the case that what we know is infinite (which happen with infinite sets, for instance, with the natural numbers, or, more interestingly, with time and space coordinates and regions).

Logical entailment – properties (P4) (same as world logics)

Definition (Monotonicity). Let w , be any formula of some logical language, T and Σ any set of formulas. Then, we have monotonic inference when, for all formulas w ,

$$\text{If } T \models w \text{ then } T \cup \Sigma \models w$$

Observation (Monotonicity) Monotonicity implements a fundamental and intuitive property of knowledge, for instance of **scientific knowledge**. If knowledge increases then what can be derived from it via entailment can only increase.

At worst, the new knowledge will not change anything if the new piece of knowledge was implied by what is already known.

Definition (monotonic logics). Monotonic logics are logics which allow only for monotonic inference

Logical entailment – properties (P5) (same as world logics)

Definition (NonMonotonicity). Let w , be any formula of some logical language, T and Σ any set of formulas. Then, we have nonmonotonic inference if there exists a formula w such that

$$T \models w \text{ and } T \cup \Sigma \not\models w$$

Observation (NonMonotonicity) Monotonicity is a property which sometimes does not hold. This is extensively the case with common sense reasoning, a topic extensively studied in AI.

Example (Historical AI). the belief that all birds fly can be defeated by the fact that penguins are birds and they do not fly.

Example (Historical scientific knowledge). The discovery that it is the earth rotating around the sun.

Example (Everyday life example). The world evolve, and a new perception may invalidate previous knowledge

Definition (NonMonotonic logics). Monotonic logics are logics which allow for both monotonic and nonmonotonic inference.

Practical point of view: the logics used in mathematical reasoning and in formal methods, as applied to, e.g., programming languages, are monotonic, while most logics defined in AI are nonmonotonic. Negation by failure, as implemented in relational DBs is nonmonotonic.

Logical entailment properties (P6)

Deduction theorem (Logical consequence, validity), with T possibly empty:

$$T, \phi \models \psi \text{ if and only if } T \models \phi \supset \psi$$

Observation 1: The deduction theorem explains (left to right) the meaning of implication. Implication is how we express logical consequence in language.

Observation 2: It also says (right to left) that from absurdity (i.e, $P \wedge \neg P$), we can derive everything, any formula (and assertion) A.

Logical entailment properties (P7)

Refutation principle (Logical consequence, unsatisfiability), with T possibly empty:

$T \models \phi$ if and only if $T \cup \{\neg \phi\}$ is unsatisfiable

Observation 1: The refutation principle explains the meaning of negation. It captures the fact that absurdity (i.e, $P \wedge \neg P$) cannot be satisfied by any model depicting facts in the real world.

Observation 2: Algorithmmitically, it suggests how to reason backwards from goals.

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Reasoning as entailment

Reasoning Problem (Model checking). Given a theory T and a model M , check whether $M \models T$.

Reasoning Problem (Satisfiability). Given a theory T , check whether there exists a model M such that $M \models T$.

Reasoning Problem (Validity). Given a theory T , check whether for all models M , $M \models T$.

Reasoning Problem (Unsatisfiability). Given a theory T , check whether there is no model M such that $M \models T$.

Reasoning Problem (Logical consequence). Given T_1 and T_2 , check whether $T_1 \models T_2$;

Reasoning Problem (Logical equivalence). Given T_1 and T_2 , check whether $T_1 \models T_2$ and $T_2 \models T_1$.

Example: Valid, Satisfiable or Unsatisfiable?

Prove that

- **Blue** Formulas are valid,
- **Magenta** Formulas are satisfiable but not valid
- **Red** Formulas are unsatisfiable.

Satisfiable

Unsatisfiable

$$A \supset A$$

$$A \vee \neg A$$

$$\neg \neg A \equiv A$$

$$\neg (A \wedge \neg A)$$

$$A \wedge B \supset A$$

$$A \supset A \vee B$$

$$A \vee B$$

$$A \supset B$$

$$\neg (A \vee B) \supset C$$

$$A \wedge \neg A$$

$$\neg (A \supset A)$$

$$A \equiv \neg A$$

$$\neg (A \equiv A)$$

Valid

Not Valid

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Model Checking

Given a theory T and a model M , check whether $M \models T$.

For example we can determine whether

$$(\neg p \vee q) \wedge (q \supset \neg r \wedge \neg p) \wedge (p \vee r)$$

is a model for

$$p = T, q = F, r = T \quad \text{or} \quad p = F, q = F, r = F.$$

p	q	r	$\neg p \vee q$	$\neg r \wedge \neg p$	$q \supset \neg r \wedge \neg p$	$p \vee r$	Answer
T	F	T	F	F	T	T	F
F	F	F	F	T	T	F	F

Observation. This is useful for checking properties (the input theory T) of existing (artificial or natural) systems (the model M).

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Satisfiability

Given a theory T , check whether there exists a model M such that $M \models T$.
For example, we can determine if

$$(\neg p \vee q) \wedge (q \supset \neg r \wedge \neg p) \wedge (p \vee r)$$

is satisfiable.

p	q	r	$\neg p \vee q$	$\neg r \wedge \neg p$	$q \supset \neg r \wedge \neg p$	$p \vee r$	Answer
T	T	T	T	F	F	T	F
T	T	F	T	F	F	T	F
...	...	...	...	...	...	...	...
F	F	T	T	F	T	T	T
F	F	F	T	T	T	F	F

Observation. The first reasoning problem by excellence! Given a set of requirements (the theory T) find a model which satisfies it (e.g. TSP, scheduling)

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Validity

Given a theory T , check whether there for all models M we have $M \models T$.

For example, we can determine if

$$(p \supset q) \vee (p \supset \neg q)$$

is a valid formula or not.

p	q	$p \supset q$	$\neg q$	$p \supset \neg q$	$(p \supset q) \vee (p \supset \neg q)$
T	T	T	F	F	T
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

Observation: Find whether a property is true in all models (of interest).
Useful for theory reformulation (using, e.g., equivalence)

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Unsatisfiability

Given a theory T , check whether there is no model M such that $M \models T$.

For example, we can determine if

$$\neg((p \supset q) \vee (p \supset \neg q))$$

is unsatisfiable or not.

p	q	$p \supset q$	$\neg q$	$p \supset \neg q$	$\neg((p \supset q) \vee (p \supset \neg q))$
T	T	T	F	F	F
T	F	F	T	T	F
F	T	T	F	T	F
F	F	T	T	T	F

Observation: Find whether a property (the theory T) is not realisable. Useful to evaluate the suitability of a LoDE theory (e.g., AI, non monotonic reasoning, planning).

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Logical Consequence

Given two theories T_1 and T_2 , check whether $T_1 \models T_2$.

For example, we can determine if

$\neg q \vee \neg p$ is a logical consequence of the formula $\neg q$.

p	q	$\neg p$	$\neg q$	$\neg q \vee \neg p$
T	T	F	F	F
T	F	F	T	T
F	T	T	F	T
F	F	T	T	T

Whenever $\neg q$ is True,
 $\neg q \vee \neg p$ is also True,
making it a logical
consequence of $\neg q$.

Observation: The second reasoning problem by excellence. Compute the consequences of a set of facts.

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Logical Equivalence

Given two theories T_1 and T_2 , check whether $T_1 \models T_2$ and $T_1 \models T_2$.

For example, using the truth table method we can determine whether $p \supset (q \wedge \neg q)$ and $\neg p$ are logically equivalent.

p	q	$q \wedge \neg q$	$p \supset (q \wedge \neg q)$	$\neg p$
T	T	F	F	F
T	F	F	F	F
F	T	F	T	T
F	F	F	T	T

The truth value is the same for every interpretation, therefore the formulas are logically equivalent.

Observation: Useful to substitute equivalents for equivalents (property reformulation).

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Reasoning problems - Correlations

Theorem. If a formula is valid, then it is also satisfiable, and it is also not unsatisfiable. That is:

Validity implies **Satisfiability** equivalent to **not Unsatisfiability**

Theorem. If a formula is unsatisfiable, then it is also not satisfiable, and also not valid. That is:

Unsatisfiability equivalent to **not Satisfiable** implies **not Valid**

Reasoning Problem correlations

Observation 1. Differently from Satisfiability, testing the holding of Validity or Unsatisfiability requires checking all the 2^n interpretations for success. With satisfiability this is only a worst case analysis (only one model, which is also the last to be selected).

Observation 2. For any finite set of formulas T , (i.e., $T = A_1, \dots, A_n$ for some $n \geq 1$), T is valid (respectively, satisfiable and unsatisfiable) if and only if $A_1 \wedge \dots \wedge A_n$ (respectively, satisfiable and unsatisfiable)

Observation 3. All mainstream reasoning algorithms implement SAT and, to a lesser extent, UNSAT.

Reasoning problems - Correlations

Theorem. The validity, satisfiability and unsatisfiability of a formula and of its negation correlate as follows:

If A is	then $\neg A$ is
Valid	Unsatisfiable
Satisfiable	Not Valid
Not Valid	Satisfiable
Unsatisfiable	Valid

Reasoning problems - Correlations

Model checking (= entailment) (MC) is the core decision problem

Satisfiability (SAT) reduces to success in proving MC in (at least) one model

Validity (VAL) reduces to success in proving MC for all models

Unsatisfiability (UNSAT) reduces to failure in proving MC in all models

Reasoning problems – Correlations

Logical Consequence (LC). Two possibilities

- Use the deduction theorem to reduce LC to a VAL problem
- Use the refutation principle to reduce to an UNSAT problem

Logical Equivalence (LE) reduces to LC.

Reasoning Problems - observations

Observation (Truth table method). Truth tables are an effective method for deciding any of the LoP reasoning problems. They are used, often manually, to check some simple problems. As presented before, they do not scale to complex problems. The DPLL decision procedure is their concrete implementation, the state of the art of LoP reasoning in real world problems.

Observation (Model generation). All the problems described above are based on the same steps: (1) generate all the interpretations, (2) for each interpretation perform model checking, that is, compute whether it is a model for the input, possibly via the intermediate computation of some sub-formulas of the input, and (3) decide whether the answer to the input problem is YES/NO.

Observation (Deciding SAT /UNSAT/ VAL). The process for these three problems is the same. The termination condition is different. With SAT you terminate as soon as you find an interpretation which is a model. With VAL and /UNSAT you need to exhaustively check all interpretations.

Observation (Deciding LC / LE). With LOC /LE you need to exhaustively check all interpretations. With LE you need always to compare the truth values of the two components of the equivalence. With LC you need to check the second component only when the first evaluates to True.

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Key notions

- LoP facts, LoP percepts
- Negation, disjointness, equivalence, conjunction, disjunction, implication
- Complete model, minimal model
- Model as interpretation
- Partiality as set of models
- Three step model building
- Model checking, SAT, UNSAT, VAL
- Logical consequence, logical equivalence
- Deduction theorem
- Refutation principle



LoP- The Logic of Propositions

Reasoning about what is True and what is False (HP2T)